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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 516 — Past Questions & Materials
by Renance

Course: PEE 516
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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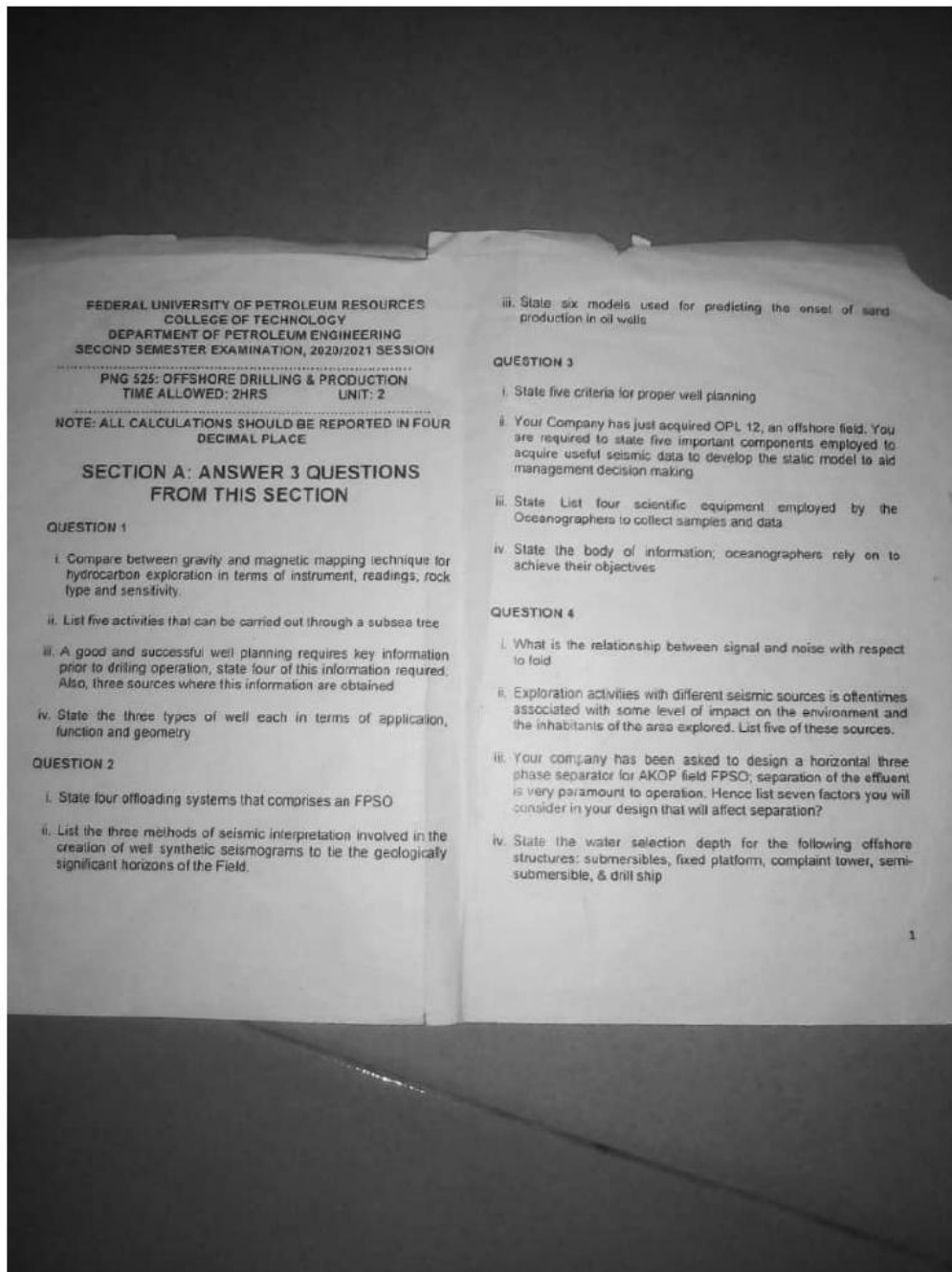
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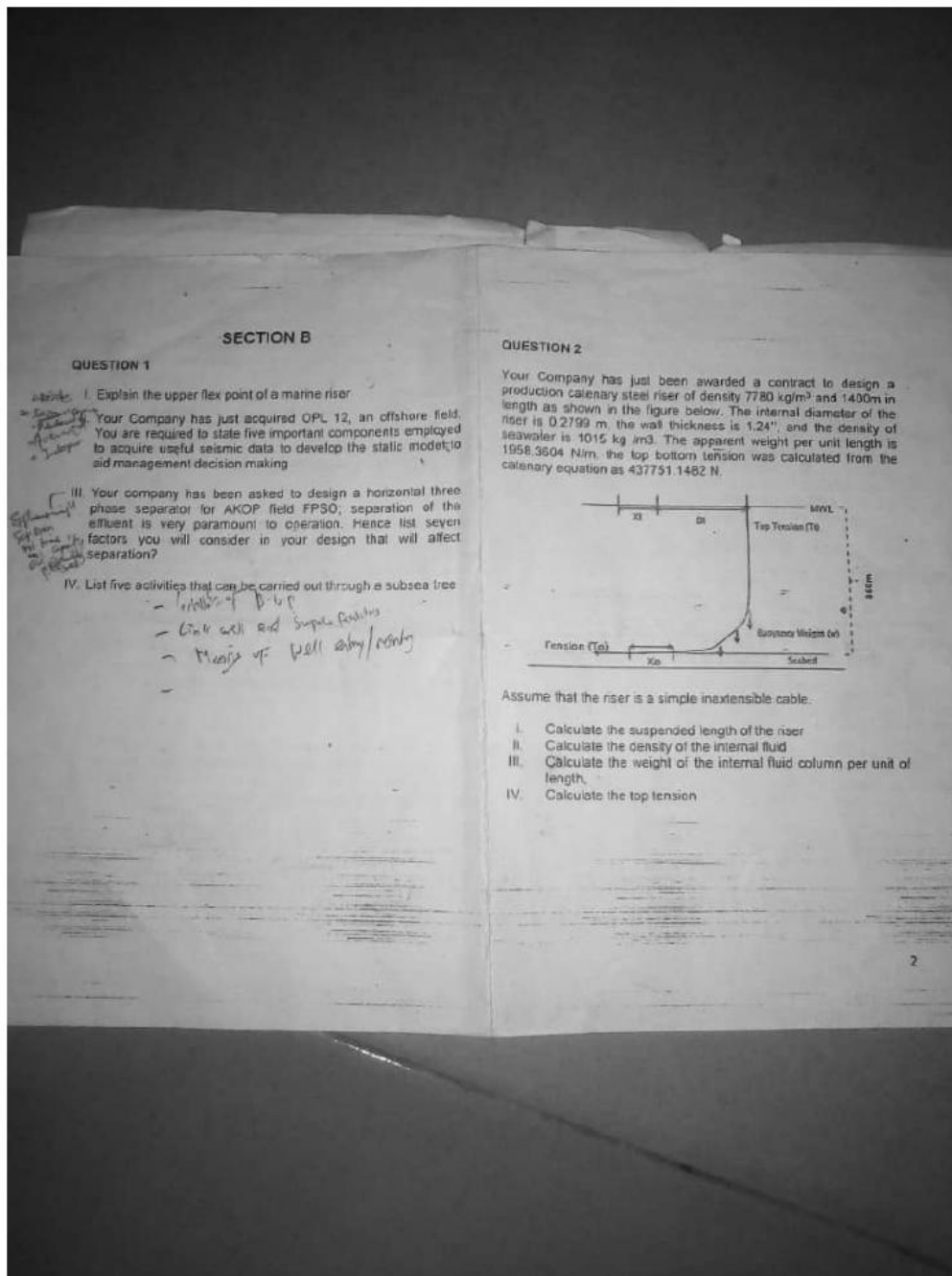
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Paper 1 of 2 — PEE 516 PQ (Offshore Drilling and Production)

Level 500 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2019/2020 SESSION
PNG 525: OFFSHORE DRILLING & PRODUCTION
TIME ALLOWED: 2HRS UNIT: 2
INSTRUCTIONS: ANSWER ALL QUESTIONS IN SECTION
A & ONE FROM SECTION
NOTE: ALL CALCULATIONS SHOULD BE REPORTED IN
FOUR DECIMAL PLACE

SECTION A

QUESTION 1

Use the well log data in the table below and the quick look approach proposed by Tixeront's rule in predicting sand production. Determine if there will be sand production from the depth given.

Depth (ft)	Matrix Shear Time (mic s/ft)	Matrix Compression time (mic s/ft)	log Compression time (mic s/ft)	log Shear time (mic s/ft)	Rock Matrix Density (gm/cm ³)	log Bulk density (gm/cm ³)
1000	50	85	78	127	2.62	2.24
2000	56	87	89	139	2.76	2.33

QUESTION 2

- Explain the function of the three distinct sections of Truss Spar.
- List the three methods of seismic interpretation involved in the creation of well synthetic seismograms to tie the geologically significant horizons of the Field.
- State six models used for predicting the onset of sand production in oil wells.
- Often time, sand is usually produced in newly drilled or old wells. State five possible causes of sand production.

QUESTION 3

A proposed field development at FUPRE is required to incorporate a water disposal facility whose separation and treatment plant produces 9200 BPD of water containing 2400mg of oil per liter of water with a maximum oil droplet size of 550 μ m. The water viscosity and specific gravity are 1.2 cp and 0.985 respectively, and the oil specific gravity is 0.86. It is required to treat the produced water to reduce its oil content of 980 mg/L, taking the retention time of 12 min, determine the dimension of:

- Horizontal cylindrical skimmer (hint: generate the table with diameter range of 60in to 120 in, using interval of 12)

QUESTION 4

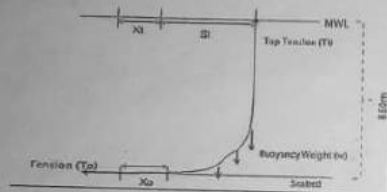
A sacrificial magnesium anode was used to protect a pipeline transporting crude oil from Escravos to Forcados terminal. It weight 150 lbs. It is designed in such a way that only half of the anode is used effectively in the protection of the pipeline. After 2 years,

- Calculate the effective weight of the anode used up.
- Also calculate the fraction of the effective life of the anode, the current generated is 2 amperes.
- Write the corrosion reaction equation and assume 100% efficiency. (At wt of Mg is 24.32gm, 1 faraday = 96500 coulombs = 26.8amp/m).

SECTION B: ANSWER ANY 2 QUESTIONS

QUESTION 5

Your Company has just been awarded a contract to design a production catenary steel riser of density 7780 kg/m^3 and 1400 mm in length as shown in the figure below. The internal diameter of the riser is 0.2793 m , the wall thickness is $1.24''$, and the density of seawater is 1015 kg/m^3 . The apparent weight per unit length is 1358.3604 N/m , the top bottom tension was calculated from the catenary equation as 437751.1482 N

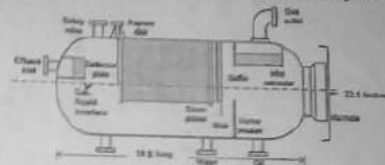


Assume that the riser is a simple inextensible cable. Calculate:

- I. The top tension
- II. The suspended length of the riser
- III. The density of the internal fluid

QUESTION 6

Given a horizontal separator in the figure below, designed to operate at a temperature of 70°F and pressure of 480 psia . It is equipped with a knitted wire mist extractor having a separation coefficient of 0.382 . The oil gravity is 35°API and a retention time of 90 seconds. The apparent molecular weight of the gas is 22.35 g/mol . Take the apparent molecular weight of air = 28.96 g/mol .



Will you consider the separator adequate assuming that the area of the gas is equal to the separator area? Give reason(s) to justify your answer.

QUESTION 7

A sacrificial magnesium anode was used to protect a pipeline transporting crude oil from Escravos to Forcados terminal. It weight 150 lbs . It is designed in such a way that only half of the anode is used effectively in the protection of the pipeline. After 2 years,

- i. Calculate the effective weight of the anode used up.
- ii. Also calculate the fraction of the effective life of the anode, the current generated is 2 amperes.
- iii. Write the corrosion reaction equation and assume 100% efficiency. At wt of Mg is 24.32 gm , $1 \text{ faraday} = 96500 \text{ coulombs} = 26.8 \text{ amp/h}$.

Paper 2 of 2 — PEE 516 (Offshore Drilling and Production)

Level 500 • First Semester • Handout

Offshore

Drilling and

Production

Offshore Drilling and

Production

BY

OKOTIE SYLVESTER

Course

Outline

- ☐ Introduction
- ☐ Offshore Exploration
- ☐ Site preparation and Field architecture
- ☐ Offshore drilling and production platforms
- ☐ Offshore structures installation
- ☐ Offshore Drilling
- ☐ Subsea Facilities
- ☐ Remote Operated Vehicle (ROV) operations
- ☐ Mooring Systems
- ☐ Production Operation
- ☐ Flow Assurance
- ☐ Offshore Environmental concern

15/02/2025 2

Offshore

Drilling and

Production

Lesson I

Course

CourseIntroduction

Introduction

Introductio

n

- ☐ The exploration and production of oil and gas has gradually moved from onshore to the offshore environment.
- ☐ Due to the depletion of onshore and offshore shallow water reserves, exploration and production operations is advancing into frontier deep water basins.

□ The oil and gas industry is currently pushing into a more hostile and remote environment (ultra-deep water) which has become a challenge to the oil and gas industry.

□ This environment requires the development of new technology in order to exploit the huge hydrocarbons deposit in this region that are inaccessible to exploit with the existing onshore or shallow water technologies.

15/02/2025 4

Offshore

Introductio

Drilling

n and

Production

□ Deep water oil drilling is so attractive to the oil companies despite the hostile environment because of the little share of the government in this area and more difficult for the government to regulate the activities of the oil companies.

□ It has fewer disturbances by communities and militants attacks on personnel and facilities and also reduces oil theft.

15/02/2025 5

Offshore

Introductio

Drilling

n and

Production

□ The offshore environment has some main constraints in its exploration and production operations;

- since it is fully remotes works with no direct human action,
- the safety issues related to intervention on “live” systems or interference with drilling operations,
- the wave, wind and current conditions and also
- the relative complexity due to the variety of subsea production systems.

15/02/2025 6

Drilling &

Production

Platform

Offshore structures have their respective unique, efficient and economic design based on some factors presented in tte next slide which are considered in the selection of such platforms.

15/02/2025 7

Drilling &
Production
Platform

The water depth selection for offshore rig selection is:

- 8 to 35 ft □ Swamp Barges
- 9 □ 150 ft □ Submersibles
- 10 □ 350 ft □ Jack-ups
- 11 □ 1,000 ft □ Fixed Platform
- 12 □ 8,000 ft □ Semi-submersible
- 13 □ 8,000 ft □ Drill Ship

15/02/2025 8

Offshore
Drilling and
Production

Factors considered in the selection of offshore platforms

- The depth of the water or ocean
- Environmental criteria (wind, wave, icebergs, storms and current movement)
- Type and density of sea water
- The depth from the seabed to the target zone (Drilling Depth)
- Operational limitations
- Type of well
- The weight, size etc of the topside equipment
- Construction constraint (space availability)
- The movement of drilling rig from one location to another (Rig move)
- Economic analysis (cost)

15/02/2025 9

Historical
Background

- Petroleum has been known throughout historical time. It was used in mortar, for coating walls and boat hulls, and as a fire weapon in defensive warfare.
- Native Americans used it for magic, medicine and paints making.
- Pioneers bought it from the Native Americans for medicinal use and called it Seneca oil and Genesee oil.
- In Europe it was scooped from streams or holes in the ground. In the early 19th century, small quantities were made from shale and also, several streets in Prague in 1815 were lighted with petroleum lamps.

15/02/2025 10

Historical

Background

- The offshore exploration of oil and gas dates back to the nineteenth century.
- The first offshore oil wells were drilled from extended piers into the waters of Pacific Ocean, offshore Summerland, California in the 1890s (and offshore Baku, Azerbaijan in the Caspian Sea).
- However, the birth of the offshore industry is commonly considered as in 1947 when Kerr-McGee completed the first successful offshore well in the Gulf of Mexico in 15 ft (4.6 m) of water off Louisiana with a fixed platform.
- Around 1891, the first submerged oil wells were drilled from platforms in Ohio.

15/02/2025 11

Historical

Background

Since the installation of this first platform in the Gulf of Mexico over 80 years ago, the offshore industry has seen many innovative structures (fixed and floating), placed in progressively deeper waters and in more challenging and hostile environments.

In Nigeria, the first offshore field, Bonga was discovered in 1996 and is currently being operated by Shell Nigeria who own 55% of the license in collaboration with Exxon 20%, Nigeria Agip 12.5% and Elf Petroleum Nigeria Limited 12.5%.

Production started in November 2005 with current production of 202,000 barrels per day and estimated oil in place of 1470000 MMbbl due to the addition of second block (Bonga SW) in May 2001 and third block (Bonga North) in 2004.

15/02/2025 12

Historical

Background

- The second deepwater oil field (Agbami) was discovered in late 1998.
- The field operator of the field is Star Deep Water Limited, an affiliate of Chevron.
- The field partners alongside Chevron are: Famfa oil, Petrobras, Statoil and NNPC.
- Production started in 2008 at over 70,000 barrels per day with peak production estimate to be at approximately 250,000 bbls/d with an

estimated oil place of 900MMbbl.

15/02/2025 13

Historical

Background

- There several other offshore field on stream now such as Usan discovered in 2002 and operated by Total E&P Nigeria.
- Another is Total Egina field whose development is currently going on and production scheduled to start in 2018. The estimated oil in place is 550 MMbbl and currently the FPSO is currently under construction in Korea.
- We have Akpo, Era, Aje, Abo etc.

15/02/2025 14

Main

Offshore

building

blocks

- The architecture of offshore fields is far more complex than onshore fields because; majority of the installations are laid on the seabed.
- Drilling and production operations in offshore environment also require some facilities such as risers to exclude the water body, mooring lines, loading buoy etc.

15/02/2025 15

Main

Offshore

building

Key areas of Offshore Field Engineering

blocks

Well

system

Flow Subsea

assurance structures

Main Pipeline,

System

Building flowlines

design

Blocks & risers

Underwater Controls &

operations Umbilicals

Offshore

Operations

15/02/2025 16

Main

Offshore

Well system

building

blocks

Drilling downhole

Xmas tree

completion casing

Workover

Well head Riser

B.O.P

WELL SYSTEM

15/02/2025 17

Main

Offshore

building

Pipelines, Flowlines & Risers

blocks

Export pipelines pigging risers

Inter-field flowlines

risers Tie-ins ESD valves

PIPELINES, FLOWLINES &

RISERS

15/02/2025 18

Main

Offshore

building

Subsea structure blocks

Pipe and valve system manifolds

Protective structures Templates or clusters

SUBSEA STRUCTURE

15/02/2025 19

Main

Offshore

building

Subsea Controls & Umbilicals

blocks

Control umbilicals & chemical

injection

Topide control Subsea control modules and

equipment sensors

SUBSEA CONTROLS &

UMBILICALS

15/02/2025 20

Main

Offshore

building

Offshore operations blocks

Land testing & offhore

installations

Hook-up & commisioning

start-up Pre & Post seabed surveys

OFFSHORE OPERATIONS

15/02/2025 21

Main

Offshore

building

Underwater Operations

blocks

diving operations

Installations

Robots

Inspection, maintenance & repair

UNDERWATER OPERATIONS (IMR)

Remote operated vehicle

(ROV)

15/02/2025 22

Main

Offshore

building

Field & Reservoir Operations

blocksField &

Reservoir

Site

layout-

drilling-
data
Downhole
completion
Reservoir
engineering
Surface/subsurface
production
engineering (process)
Water, gas & chemical
injection
Artificial lift system (gas lift &
electrical submersible pump
[ESP])
logistics (supply of tools and
spares)
Ease of maintenance & workover
operations
Government requirements
15/02/2025 23
Main
Offshore
building
Project & Field Management blocks
Project &
Management
Costs &
controls
Planning &
schedule
QA/QC
certification
Procurement
Government
requirements
Construction/Test
Installations
Partnership
Project execution
Project monitoring

15/02/2025 24

Offshore

Drilling and

Production

15/02/2025 25

Offshore

Drilling and

Production

Offshore Drilling and

Production (PNG 525)

BY

OKOTIE SYLVESTER

Course

Outline

☐ Introduction

☐ Offshore Exploration

☐ Site preparation and Field architecture

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☐ Production Operation

☐ Flow Assurance

☐ Offshore Environmental concern

15/02/2025 2

Offshore

Drilling and

Production

Lesson III

Site

Site Preparation

Preparation and

and Field

Field Architecture

Architecture

Site

preparation

□ In offshore production, oil and gas are extracted from the wells and are brought to the surface production facilities via the riser system above the ocean surface or seabed.

□ The selection of a particular type of platform, pipelines and vessels that will best suit the region to drilling and produce the oil and gas depends on the field's location, water depth, climate and the facility's size and capabilities.

□ Besides, the environmental, safety, stakeholder and financial issues are also considered when selecting among the options.

15/02/2025 4

Site

preparation

□ The design and installation of offshore structures cannot be successfully completed without the expertise of oceanographer.

□ Some reservoirs are located in hurricane-prone regions; which implies that the facilities to be installed in such area need to be specifically designed to resist extreme waves, winds and also to minimize environmental damage.

□ While some are in areas where there are intense ground movements caused by earthquake.

15/02/2025 5

Site

preparation

□ Some offshore field environment; experience massive wind, wave and also changing motions of tides and currents which can cause severe structural stress to platforms.

□ If a careful and close monitoring of the weather is not carried out to allow platform crews to prepare for evacuation by halting drilling activities, securing the facility and shutting down production, will lead to unforeseen accidents; that can culminate into loss of life and facilities.

15/02/2025 6

Why the service of

Oceanographers

□ Oceanographers use their knowledge of biology, chemistry, physics and geology to study the seas and oceans.

□ They carry out research to investigate the effects of climate change or explore the impact of pollution and offshore engineering on marine life.

□ They study the marine environment with observational skills to collect data and track changes in the terrain closely; to know when certain types of aquatic animals migrate in or out of the terrain.

□ If this aquatic lifestyle is known, allowance maybe created to shutdown

operations in that period of time.

□ They also study the movement of iceberg through a combination of actual ice force measurements, ice model tests and engineering analyses.

15/02/2025 7

Why the service of
Oceanographers

□ Oceanographers use the following branches of knowledge to achieve their objectives:

- Biological – studying marine plants and animals.
- Physical – exploring water temperature, density, wave motion, tides and currents.
- Geological – examining the structure and make-up of the ocean floor.
- Chemical – analysing the chemicals in sea water and the impact of pollutants.

15/02/2025 8

Why the service of
Oceanographers

Oceanographers use a variety of scientific equipment to collect samples and data, including:

- Remote sensors on satellites
- Instruments on towed or self-powered underwater vehicles
- Scientific apparatus like sensors on moored or drifting buoys
- Probes lowered into the sea
- Drills to collect sediment cores from the seabed
- Microphones to measure acoustics
- Marine robots to explore the seabed

15/02/2025 9

Field
Architecture

Offshore fields architectural designs differs considerably from one field to another, this implies that no two fields producing oil and/or gas have the same architectural configuration

such as the hydrocarbon reservoir, well requirements, drilling and completion, process support (platform), subsea storage and facilities layout (Figure in next slide) etc

but the same building /sub-building blocks are used. In an offshore (submerged) oil well, space for storing equipment and all infrastructural needs are built directly into the oil platform.

15/02/2025 10

Field

Architecture

15/02/2025 11

Field

Architecture

Offshore field development can either be a stand-alone or tie back to a fixed platform or tension leg platform (TLP) or float production storage and offloading (FPSO) platform.

15/02/2025 12

Field

Architecture

Consideration for subsea field development

In subsea field development, the following issues should be considered:

- ☐ Is the proposed field be to developed located in deep water or shallow-water;
- ☐ Should dry tree on the surface of the platform or wet tree on the seabed be installed;
- ☐ Is there a new constructed production facility (Stand-alone) or tie-back development to existing field;
- ☐ Subsea processing (where will the gas be sent to); Hydraulic and chemical units;
- ☐ Artificial lift methods;
- ☐ Facility configurations (i.e., template, well cluster, satellite wells, manifolds).

15/02/2025 13

Field

Architecture

Key Factors Guiding/ Dictating the Architecture Design

- ☐ What is the depth of the reservoir and the corresponding well paths or trajectories?
- ☐ The design of the entire production system and related flow assurance issues
- ☐ Pre drilling assessments of geological and environment state that may lead to damage or risk (Geohazards) and the study of the underwater depth of the ocean floor (bathymetry)
- ☐ Drilling rig to use and the pattern of mooring system
- ☐ Dominating Weather and Environmental Conditions

15/02/2025 14

Field

Architecture

Key Factors Guiding/ Dictating the Architecture Design

- Supply Boat Movements, Loading / Offloading / Dropped Objects
- Optimal Location of Subsea Facilities (the submerged production system {SPS} + the umbilical, riser and flowline {URF} systems) & Access to wells
- Permanent production facilities mooring lines
- FPSO weather vanning and shuttle tanker offloading
- Other existing Facilities in the vicinity

15/02/2025 15

Field

Architecture

15/02/2025 16

Offshore

Drilling and

Production

Offshore Drilling and

Production

BY

OKOTIE SYLVESTER

Course

Outline

- Introduction
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- Flow Assurance
- Offshore Environmental concern

15/02/2025 2c

Offshore

Drilling and

Production

Lesson VI

Offshore

Offshore Drilling

Drilling

Introduction

n

□ Offshore drilling is a mechanical process where a wellbore is drilled into the seabed.

□ It is typically carried out to explore for hydrocarbon; and

□ Subsequently extract petroleum which lies in rock formation beneath the seabed.

□ Prior to drilling the well, the guard, wellhead and BOP are secured in place and the following activities take place:

15/02/2025 4

Drilling

Steps

□ Spud in, a process of beginning to drill a well or to start the well drilling process by removing rock dirt and other sedimentary with the drill bit.

□ Drill ahead

□ Drill to first casing

□ Pull out of hole (trip)

□ Run first casing – 30” conductor

□ Pump cement

□ Displace cement behind casing

□ Drill ahead to second casing seat

□ Run second casing string -18/3/4” inner conductor (with wellhead welded to top)

□ Pump and displace cement & install BOP on wellhead

15/02/2025 5

Drilling

Steps

□ Next – drill – case – cement approaching reservoir depth.

□ Drill ahead approaching reservoir

□ Drill to total depth (TD)

□ Pull out of hole and run logs

□ Run production liner – 7” and

□ Cement

□ Suspend well and remove bop

□ Run production tubing (51/2”),

□ Run tree, connect tree to flowlines,

□ Perforate casing and flow the well

15/02/2025 6

Drilling rig floor
layout and
components

1. Mud tank
2. Shale shakers
3. Suction line(mud pump)
4. Mud pump
5. Motor or power source
6. Vibrating hose
7. Draw-works
8. Standpipe
9. Kelly hose
10. Goose-neck
11. Traveling block
12. Drill line
13. Crown block
14. Derrick
15. Racking Board (Sometimes referred to as the Monkey Board)

15/02/2025 7

Drilling rig floor
layout and
components

16. Stand(of drill pipe)
17. Setback(floor)
18. Swivel(On newer rigs this may be replaced by a top drive)
19. Kelly drive
20. Rotary table
21. Drill floor
22. Bell nipple
23. Blowout preventer(BOP) Annular type
24. Blowout preventer(BOP) Pipe ram & blind ram
25. Drill string
26. Drill bit
27. Casing head or Wellhead
28. Flow line

15/02/2025 8

Drilling rig floor
layout and
components

Derrick

Derrick is the support structure for the equipment such as block and tackle used to lower and raise the drill string and casing into and out of the wellbore. It carries the hook load, allows the movement of equipment on the rig floor and support the top drive (See Figure in the previous slide).

Rotary table

The principal component of a rotary, or rotary machine, used to turn the drill stem and support the drilling assembly. It has a bevelled gear arrangement to create the rotational motion and an opening into which bushings are fitted to drive and support the drilling assembly.

15/02/2025 9

Drilling rig floor
layout and
components

Drawworks

It is the hoisting mechanism on a drilling rig. Simply puts, it is the mechanical section made up of a large winch that contains the spool, whose main function is to reel in/out the drill line to raise/lower the traveling block and thus raises or lowers the drill stem and bit.

Drill string

Is an assembled collection of drill pipe, heavy weight drill pipe, drill collars and any of a whole assortment of tools, connected and run into the wellbore to facilitate the drilling of a well. The collection is referred to singularly as the drill string.

15/02/2025 10

Drilling rig floor
layout and
components

Mousehole

A Mousehole is a shallow bores under the rig floor, usually lined with pipe, in which joints of drill pipe are temporarily suspended for later connection to the drill string.

Rat-hole

A hole in the rig floor 30 to 35 feet deep, lined with casing that projects above the floor. The Kelly is placed in the rathole when hoisting operations are in progress.

15/02/2025 11

Drilling rig floor

layout and
components

Slips

- Wedge-shaped pieces of metal with teeth or other gripping elements that are used to prevent pipe from slipping down into the hole or to hold pipe in place.
- Rotary slips fit around the drill pipe and wedge against the master bushing to support the pipe.
- Power slips are pneumatically or hydraulically actuated devices that allow the crew to dispense with the manual handling of slips when making a connection.
- Packers and other down hole equipment are secured in position by slips that engage the pipe by action directed at the surface.

15/02/2025 12

Drilling rig floor
layout and
components

Hydraulic operated power slip

- Hydraulic operated power slips are slips equipped with replaceable slips and insert carriers to handle various styles and sizes of tubular.
- They can handle casing, drill pipe, drill collars and tubing.
- They are used in combination with the Rotary Support Table.

15/02/2025 13

Offshore
Drilling and
Production

15/02/2025 14

Offshore
Drilling and
Production
Offshore Drilling and
Production

BY
OKOTIE SYLVESTER

Offshore
Drilling and
Production

Lesson X
Production
Production Operation
Operation

Production

Operation

- Production engineers, including subsurface engineers, manage the interface between the reservoir and the well, including:
- Perforations, sand control, downhole flow control, and downhole monitoring equipment; evaluate artificial lift methods; and
- Also select surface equipment that separates the produced fluids (oil, natural gas, and water).
- Production of oil and gas offshore is similar to production onshore with the following major differences:

15/02/2025 3

Production

Operation

□ Concept Selection for Field Development

– To define likely development schemes for a given field, one has to match production facilities with the field requirements.

□ The field requirements that drive concept selection include:

□ production volumes,

□ environment, Waves, current, (Spread or weather vane moored FPSO, buoy or tandem loading mode)

□ water depth, (platform, FPSO etc)

□ distance to shore or infrastructure,

□ Reservoir Characteristics (large reservoir developed with subsea installations, small reservoir developed with dry well heads)

□ the number of drilling centers required to drain the reservoir,

□ reservoir fluid and the well intervention (work-over) frequency,

□ The risk to personnel.

15/02/2025 4

Production

Operation

□ Surface production operations involve mainly fields processing of fluids as they emerge from the wellhead.

□ The process involves mass, heat and momentum transfer treatment in separator units that are necessary to produce high quality fluids to:

□ Meet market specifications

□ Meet statutory requirements

□ Fluid re-injection requirements such as produced water re-injection, etc.

15/02/2025 5

Production

Operation

□ Processing Techniques Involve:

- Separation: Separation of oil, gas and water phases
- Dehydration: Removal of basic sediments and water (B,S&W) from the crude
- Desalting: Reduction of salt content in the crude and water
- Sweetening: Removal of hydrogen sulphide and other sulphur contents
- Stabilization: Removal of the most volatile components from the hydrocarbons.

15/02/2025 6

Production

Operation

15/02/2025 7

Production

Operation

□ Separation Section

□ The effluent coming from the well is separated using a separator which can be vertical or horizontal or spherical.

□ A separator is a pressure vessel for separating a well stream into gaseous and liquid components.

□ They are installed on offshore and onshore facilities.

15/02/2025 8

Production

Operation

□ Functions of a Separator

The functions of a separator are as follows:

- Perform primary phase separation.
- Ensure that gas & liquid flow rate are low enough for gravity segregation & vapor-liquid equilibrium.
- Minimize turbulence & reduce velocity.
- Eliminate re-entrainment of separated gas & liquids.
- Provide outlet for gas with suitable controls to maintain preset operating pressure.
- Process gas to remove condensable water vapor that might cause hydrate formation.

15/02/2025 9

Production

Operation

□ Principles of Separator's Operation

15/02/2025 10

Production

Operation

□ Definition of some components:

- Inlet device located in pre-separation zone/section for preliminary phase separation;
- Baffles downstream the inlet component to improve flow distribution;
- Separation enhancement device located in the primary separation (gravity settling) section for major phase separation;
- Mist extraction device located in gas space to further reduce liquid content in the bulk gas stream;
- Various weirs to control the liquid level or interface level;
- Vortex breaker to prevent gas carry under at outlet of liquid phase;

15/02/2025 11

Production

Operation

□ Principles of Separator's Operation

- Fluid enters through inlet and hits inlet diverter, causing change in momentum.
- Initial gross separation of liquid & vapour occurs at the inlet divert.
- Liquid droplets, drop out of gas stream to the bottom of vessel due to force of gravity.
- Retention time required to let entrained gas evolve.
- Also, provide surge volume if necessary to handle intermittent slugs of liquid.
- Liquid leaves the vessel through liquid dump valve.
- Liquid dump valve is regulated by level controller

15/02/2025 12

Production

Operation

□ Principles of Separator's Operation

- Gas flows over inlet diverter & then horizontal through gravity settling section.
- Small liquid droplets entrained in gas are separated by gravity & fall to gas-liquid interface.
- Gas evolves through mist extractor.
- Lest liquid droplets are finally extracted at mist extractor.
- Pressure controller, controls the pressure in the separator.
- If pressure controller senses changes in pressure, it sends a signal to either open or close pressure control valve.

15/02/2025 13

Production

Operation

Factors Affecting Separation

- Gas & liquid flow rate.
- Equipment Design/Configuration.
- Operating temperature.
- Presence of impurities such as
- Operating pressure.
- Composition of well effluent. □ Paraffin
- Surge or slugging tendencies of feed □ Sand streams. □ Scale. etc.
- Physical properties of the fluid, like □ Foaming tendencies of the crude oil.
- Density, □ Corrosive tendencies of the liquid or
- Compressibility, etc. gas.
- Designed degree of separation □ Interface area between liquid & gas.
- (e.g removing 100% of particles > than 10 microns.)

15/02/2025 14

Production

Operation

□ Classification of Separators

- We have four types of classification, namely;
- Classification according to configuration.
- Classification according to number of phases.
- Classification according to duty.
- Classification according to position in train.

□ Classification according to configuration

- We have three types, namely;
- Vertical Separator.
- Horizontal Separator.

□ Single Tube Horizontal Separator.

□ Double Tube Horizontal Separator.

- Spherical.

15/02/2025 15

Production

Operation

□ Classification according to number of phases

- The classification is grouped into two:
- Two-phase separators. (separate oil from gas)
- Three-phase separators. (separate oil from gas, water)

☐ Classification according to duty

- Production separator.
- Test separator.
- Metering separator

15/02/2025 16

Production

Operation

☐ Classification according to position in train

- First stage
- Second stage
- Third stage
- High pressure
- Medium pressure
- Low pressure

☐ Separator Design/Sizing

- A typical separator design must have,
- Liquid capacity design,
- Gas capacity design/ Mist extractor design,
- Retention Time
- Other internal component mentioned before,

15/02/2025 17

Production

Operation

15/02/2025 18

Production

Operation

15/02/2025 19

Production

Operation

15/02/2025 20

Production

Operation

☐ Separator Design Calculation

- A horizontal single - tube separator which has inside diameter of 24 inches and 15ft long, operating @ 400psia (Z factor = 0.909) is equipped with a knitted were mist extractor having a separation coefficient of 0.382. The oil gravity is 35 OAPI, the gas gravity is 0.70 and operating temperature is 60 OF.
- (a) If the retention time is 1minute, what is the oil capacity of the separator?
- (b) If the gas Area is half the separator Area, what is the equivalent

diameter?

- (c) Calculate the gas capacity
- (d) What will be the gas capacity if the separator were 30ft long?

15/02/2025 21

Production

Operation

15/02/2025 22

Production

Operation

15/02/2025 23

Production

Operation

☐ Surface production operations involve mainly fields processing of fluids as they emerge from the wellhead.

☐ The process involves mass, heat and momentum transfer treatment in separator units that are necessary to produce high quality fluids to:

☐ Meet market specifications

☐ Meet statutory requirements

☐ Fluid re-injection requirements such as produced water re-injection, etc.

15/02/2025 24

Production

Operation

15/02/2025 25

Production

Operation

15/02/2025 26

Production

Operation

15/02/2025 27

Production

Operation

15/02/2025 28

Production

Operation

15/02/2025 29

Offshore

Drilling and

Production

15/02/2025 30

Offshore

Drilling and

Production

Offshore Drilling and

Production

BY

OKOTIE SYLVESTER

Offshore

Drilling and

Production

Lesson XI

Flow

Flow Assurance

Assurance

Flow

Assurance

□ Some of the most severe operational hazards are associated with the transportation of fluids.

□ When oil, water and gas simultaneously flow in a well or pipeline, many potential problems can arise

What is flow assurance?

□ Flow assurance implies guarantee of flow.

□ Flow assurance is another term for multiphase transport, which covers the transmission of oil, gas and water in the same pipeline from the reservoir to the processing plant.

15/02/2025 3

Flow

Assurance

□ It covers all methods to ensure the safe and efficient delivery of hydrocarbons from the well to the receiving facilities.

□ It can also be defined as the ability to produce fluids economically from the reservoir to the production facilities over the life of the field and in all conditions & environments.

15/02/2025 4

Flow

Assurance

□ Problems/Challenges to flow

15/02/2025 5

Flow

Assurance

Waxy Crude Oil Production and Handling

□ Waxy crude oil production involves technical means to avoid blockage in pipelines due to wax deposit formation at low temperature especially for deep water applications.

□ Today, operators are using different techniques to avoid blockage to flow such as:

- Permanent chemical injection in the pipeline flow (costly method),
- Periodic pigging of the pipeline to scrap the wax deposit which disturbs the production.

15/02/2025 6

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Deep water development

□ Paraffin precipitation and deposition in flowlines and pipelines is an issue impacting the development of deepwater subsea hydrocarbon reservoirs.

□ The buildup of paraffin deposits decreases the pipeline cross-sectional area, restricts operating capacities, and places additional strain on pumping equipment.

□ Development activities in the deepwater and frontier arena face significant challenges.

□ The colder seabed temperatures and larger pressure drops caused by longer tieback lines connecting subsea trees to a production facility are of particular concern.

□ In addition, because of the high cost of deepwater wells, fluids from different reservoirs and fields are commingled and produced through long subsea lines, enhancing the potential for fluid incompatibility.

15/02/2025 7

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Wax definition

□ Wax can be defined in a context as a long chain n-paraffin hydrocarbon that are solid at room temperature when isolated, yet soluble in the crude oil mixture at elevated temperatures.

□ It presents itself as a mix paraffin crystals and oil. It varies in quantity with each oil composition.

□ These wax crystals are typically defined as the normal paraffins ranging from C18 to C60. Precipitation of wax crystals in fluid has an impact on the rheological behavior of the oil

15/02/2025 8

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• Phenomenology

□ As the temperature of the crude drops below a critical level and/or as the low-molecular-weight hydrocarbons vaporize, the dissolved waxes begin to form insoluble crystals.

□ The deposition process involves two distinct stages: nucleation and growth.

□ Nucleation is the forming of paraffin clusters of a critical size (“nuclei”) that are stable in the hydrocarbon fluid.

□ This insoluble wax itself tends to disperse in the crude.

15/02/2025 9

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• Phenomenology

□ Wax deposition into the production system (“growth”) generally requires a “nucleating agent,” such as asphaltenes and inorganic solids.

□ The wax deposits vary in consistency from a soft mush to a hard, brittle material.

□ Paraffin deposits will be harder, if longer-chain n-paraffins are present.

□ Paraffin deposits can also contain: Asphaltenes, Resins, Gums, Fine sand, Silt, Clays, Salt & Water

15/02/2025 10

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Wax Appearance Temperature

□ The wax appearance temperature (WAT) is the temperature at which the first wax crystals start to form, on cooling of paraffin/solvent systems.

□ The wax appearance temperature (WAT) or cloud point is the temperature at which wax starts to emerge from the solution as the crude oil is cooled at the atmospheric pressure.

□ The determination of the wax appearance temperature (WAT) is extremely useful to predict organic solid deposition at a given condition.

□ Different methods can be used to determine the WAT of crude oil, such as cross polarization microscopy, viscosimetry, thermomicroscopy, fluidity point, and

differential scanning calorimetry (DSC).

15/02/2025 11

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The primary chemical parameter to establish is the critical temperature at which these wax nuclei form—the wax appearance temperature (WAT). The WAT (or “cloud point”) is highly specific to each crude. The WAT value is a function of:

- Oil composition
- Cooling rate during measurement
- Pressure
- Paraffin concentration
- Molecular mass of paraffin molecules
- Occurrence of nucleating materials such as asphaltenes, formation fines, and corrosion products
- Water/oil ratio
- Shear environment

15/02/2025 12

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Two conditions required for wax deposit

- Fluid temperature < WAT
- Wall temperature < Fluid temperature

□ Deposition will not occur if the external temperature is greater than or equal to the crude oil temperature

□ If the wall temperature is close to that of the crude oil, the deposits will be relatively insignificant (case of an insulated pipe)

□ Turbulence in the flow regime produces shearing stresses likely to tear away the wax deposits

15/02/2025 13

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Mechanisms of wax deposition

□ The predominant mechanisms proposed to describe paraffin deposition are shear dispersion and molecular dispersion.

□ Shear dispersion describes the relationship between deposition rate and shear rate.

□ Shearing of the wax molecules occurs due to the hydrodynamic drag of the flowing fluid and depends mostly on the flowrate and viscosity of the fluid.

□ Higher viscosity and low flowrates result in high wax deposition rates.

15/02/2025 14

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- Molecular diffusion describes the process by which the radial temperature gradient in the line causes a concentration gradient of dissolved paraffin components in the liquid phase.
- This concentration gradient causes paraffin to diffuse to the pipe wall, where it is assumed to deposit.
- The widely recognized transport methods contributing to wax thickness on the pipe wall are molecular diffusion of dissolved wax, particle transport of precipitated wax, and sloughing of previously deposited wax

15/02/2025 15

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Measurement of wax precipitation data

There are a few basic measurements that characterize a fluid's tendency to precipitate wax. These are:

- ☐ Wax appearance temperature or cloud point
- ☐ Wax dissolution temperature
- ☐ Pour-point temperature

15/02/2025 16

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Wax appearance temperature or cloud point

- ☐ When a liquid solution or melt is lowered to the wax appearance temperature (WAT), the wax molecules form clusters of aligned chains.
- ☐ Once these nuclei reach a critical size, they become stable and further attachment of molecules leads to growth of the crystal.
- ☐ Formation of these nuclei causes the fluid to take on a cloudy appearance, hence the name cloud point.
- ☐ This also is referred to as the wax crystallization temperature or wax appearance point (i.e Temperature at which Paraffin crystals begin to appear).

15/02/2025 17

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Wax dissolution temperature

The wax dissolution temperature is defined as the temperature at which all precipitated wax has been dissolved on heating the oil.

The experimental techniques most often used for determining wax dissolution temperature are differential-scanning calorimetry and cross-polar microscopy.

Pour-point temperature

The pour-point temperature is the lowest temperature at which the oil is mobile. This is usually identified as the stock-tank-oil gelation temperature.

15/02/2025 18

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Methods available to operators for wax prevention

When a paraffin problem develops, an operator has the following possible remediation methods to consider for wells and flowlines:

- Mechanical
- Scraping
- Coiled Tubing
- Thermal
- Hot Oiling
- Hot Water
- Continuous Application of Heat
- Chemical – Wax Solvents/Dispersants
- Batch Treatments
- Continuous Treatments
- Combination Treatments with Mechanical or Thermal Techniques

15/02/2025 19

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Methods available to operators for wax prevention

□ Mechanical removal of paraffin

- The use of wireline cutting tools and flowline scrapers is the oldest method of removing wax deposits and, where practical, should be considered.
- Scraping costs roughly the same as thermal techniques.
- It is possible with some systems to perform scraping operations while the well is producing. This allows the operator to keep production levels high.

□ Mechanical Method: Pigging

- Pigging is the primary mechanical method of removing wax buildup from the internal walls of pipelines.
- The pig cuts the wax from the pipe walls; a bypass can be set with a variable-flow pass, allowing the pig to prevent wax buildup in front.

15/02/2025 20

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Methods available to operators for wax prevention

□ Mechanical Method: Coiled tubing

- Coiled tubing with the appropriate cutters at the end also can be used for wax removal—the drawback for pipeline cleaning being the limited reach of the coiled tubing.

- For wellbore cleaning this is obviously less of a problem.

□ Thermal removal of wax

- Next to scraping, removing paraffin in well bores and flow lines by melting the deposits has the longest history of continuous use in production operations.

- Melting the wax is usually by the injection of hot oil or hot water. This technique is simple, low cost and the results are immediate.

- However, unless the temperature and time requirements for melting are recognized, inadequate treatments may eventually result in the need for extensive well service operations.

15/02/2025 21

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Methods available to operators for wax prevention

□ Thermal solutions

- Thermal insulation is used to maintain temperature above the WAT

- Thermal insulation solutions: bundles, pipe in pipe, pipe coated with insulating foam

□ Thermal removal of wax: hot oil/hot water injection

- When hot oil or hot water is injected into a well, the heat of the fluids rapidly transfers to the cooler well bore equipment and waxes in the upper sections of the well.

- As pumping of the hot treating fluid continues the heated zone progresses down the well bore removing additional deposits.

- However, the treatment will not completely clean the well unless the treating fluid is hot enough and injected long enough to reach the melting point of the hard waxes at the lower end of the deposit.

15/02/2025 22

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□ Corrosion

□ Corrosion is the deterioration of a material caused by its environment.

□ It is a spontaneous chemical reaction which results in reversion from purified metal to an ionic metallic compound.

□ The rate at which metal corrode is expressed in mils penetration per year

(mpy) and the corrosion rate is dependent on the surface area of the original material deposited & the duration of exposure time.

□ Electrochemical reaction in corrosion process requires the flow of current through a closed circuit. The circuits consist of;

- The anode (positive pole) - Losses electron by the reacting iron.
- The cathode (negative pole) - Electron is gained at this site.
- An electrolyte - A conductive solution which provides the path for current between the two poles.
- Conductor - The electro flow through the conductor (as metal)

15/02/2025 23

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□ Forms of Corrosion

- Uniform attack and/or weight loss corrosion
- Most common type of corrosion normally characterized by a chemical reaction.
- It proceeds uniformly over the entire exposed surface.
- The rate varies greatly depending on the susceptibility of the material.
- The metal becomes thinner and eventually fails

□ Oxygen Corrosion (Rusting)

- Oxygen corrosion (called Rusting) is a common type.
- Oxygen is more corrosive in salt water than fresh water.
- The dissolved oxygen is the major cause of drill pipe corrosion.
- Corrosion increases drastically when carbon dioxide or hydrogen sulphide is dissolved in oxygen-contaminated water.
- Oxygen causes severe corrosion at very low concentration(<1ppm)

15/02/2025 24

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□ Forms of Corrosion

□ Galvanic Corrosion

- Corrosion of less-resistant metal is increased while the attack of resistant material decreases.
- Less resistant metal is the anodic and the other one cathodic.
- The cathodic metal does not usually corrode in this type of couple.

□ Pitting

- A form of localized attack that results in the development of holes in the metal.
- Pitting corrosion is difficult to detect because pits are likely to be covered with corrosion products.
- Pits are so small & close together that they are mistaken for surface

roughness.

- Pitting corrosion is reported the most destructive form of corrosion.
- Most pitting failure are caused by chloride ions

15/02/2025 25

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□ Factors influencing Corrosion

- The factors can be categorized into two groups;
- Factors associated mainly with the metal.
- Factors associated mainly with the environment

□ Factors Associated Mainly with the Metal

- Effective electrode potential of the metal.
- Over voltage of hydrogen on the metal.
- Tendency of metal to form an insoluble protective film.
- Chemical and physical homogeneity of the metal surface

15/02/2025 26

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□ Factors influencing Corrosion

□ Factors Associated Mainly with Environment

- Oxygen concentration.
- Presence of H₂O (sour-gas).
- Presence of CO₂ in gas condensates, under high pressure resulting in low PH or acidity of solution.
- Relatively low resistivity of formation water.
- Nature, concentration & distribution of electrolytes such as carbonic acid due to CO₂. Presence of acids & acid generating salts
- Tendency of environment to deposit a protective film on the metal surface.
- Rate of flow of solution (high pressure well effluent) against metal.
- Temperature.
- Static or cyclic stresses.
- Contact with dissimilar solutions.
- Contact with dissimilar metals.
- Stray currents.

15/02/2025 27

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□ Detection and Measurement of Corrosion

- Various techniques used to detect & measurement corrosion include;
- Visual inspection.

- Chemical test.
- Analysis of composition of produced fluids to determine CO₂, H₂S & organic acid content.
- Analysis of formed scales, etc
 - Iron count.
- Bacteriological tests of aerobic or anaerobic bacteria.
- Electrochemical test to check surface lines, well casings or other buried structures.
- Test for pipelines.
 - Resistivity of soils.

15/02/2025 28

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☐ Detection and Measurement of Corrosion

- Tests for current flow in well casings to determine casing-potential profile.
- X-ray.
- Electrical resistance probes.
- ☐ Corrosimeter.
- ☐ Linear polarization meter.
- ☐ Galvanic probe.
- ☐ Others include;
 - Corrosion coupons.
 - Hydrogen probe.
 - Corrosion records.
 - Caliper surveys.

15/02/2025 29

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☐ Corrosion Control Techniques

- The following corrosion control techniques may be used.
 - Inhibition.
 - Chemical scavenging.
 - Use of biocides

15/02/2025 30

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☐ Calculation on Corrosion

- An offshore steel water-tank was designed to be protected by “Cathodic Protection” with the dc current applied so that the tank would have negative polarity. The tank was 7m tall; open at the top and 5m in diameter. The wall

thickness was 0.9047cm and the practical life of the tank was considered at an end when any part reached 0.4385cm in thickness. The current flow was adjusted so that 0.5807amp/m² was impressed. Accidentally, wrong connections were made and the tank was made anodic.

- The error was not notice for 3years and the blame was attributed to corrosion.

- Assuming the reaction is

$\text{Fe} \rightarrow \text{Fe}^{+} + 2\text{e}^{-}$ at 100%

- Efficiency and that 26.3amp/hr are capable of dissolving 27.93gms of iron.

(a) How much iron has dissolved in the 3 years?

(b) What was the wall thickness after 3 year?

(c) What fraction of the tanks life was used up?

- $\rho_{\text{Fe}} = 7.8\text{gm/cm}^3$, Atomic weight of steel = 55.85gm, $F = 96500$ coulombs.

- 15/02/2025 31

Flow

Assurance

□ Calculation on Corrosion

Solution

- Given,

- Wall thickness = 0.9047cm

- Current, $I = 0.5807\text{amp/m}^2$

- Valency =2

- Efficiency = 100%

- Diameter = 5 = 2.5m

- Useful wall thickness = $0.9047 - 0.4385 = 0.4662\text{cm}$

Thus, total surface area when both top and base are closed with tank probability filled.

Total surface area = Top Area + Base Area + Curved surface Area.

Where

15/02/2025 32

Flow

Assurance

15/02/2025 33

Flow

Assurance

15/02/2025 34

Flow

Assurance

15/02/2025 35

Flow

Assurance

Assignment

Read up Sand production and prediction, you
will be examine on it

15/02/2025 36

Flow

Assurance

□ Calculation on Corrosion

15/02/2025 37

Offshore

Drilling and

Production

15/02/2025 38